

Now we come to the `loop()`. The first line in `loop()`, `'digitalWrite'` sends voltage and current to the `ledPin` (pin 13) using the 'high' argument ('high' means 'on'). This turns the LED on. Then it gives a delay of 1000 milliseconds. This means that the LED remains on for 1000 milliseconds or 1 second. After that the 'low' argument (line 3) turns off the current and voltage on the `ledPin`, and again (line 4) waits for a second. These four lines will run infinitely.

Connect an LED to the board as shown in Fig. 3. Connect the positive of the LED (the longer leg) to digital pin 13 of ATmega chip and the other leg to the GND digital pin.

Once the code is written, hit the 'Play' button (or CTRL+R) to compile it. After compilation, the generated hex file has to be uploaded to the ATmega. Press File→Upload To I/O Board (or CTRL+U). The programmer section handles the burning of code into the ATmega chip. The bootloader on the ATmega kickstarts your loaded program. The result is a blinking LED.

Taking it forward

The example is a simple demonstration of physical computing, though we did not use any input. You can now experiment with various sensors like the light dependent resistor (LDR) and say, make a program that dims the LED if the LDR is subjected to less light, and vice versa.

Lots of interesting combinations of sensors and outputs, actions and reactions and a whole gamut of interactive devices can be built. The references, documentation and forums are avail-

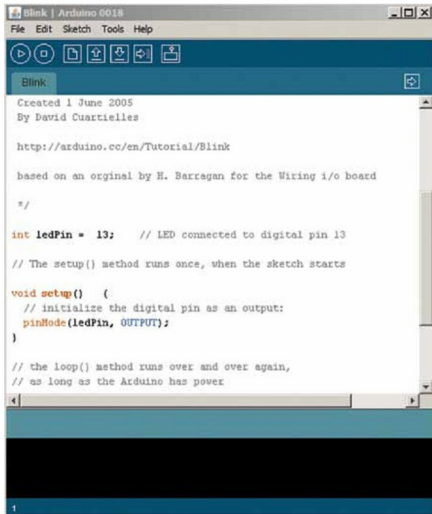


Fig. 4: The Arduino IDE

able to help you out if you run into any issues—Open Source Community! People have used this in robotics, aeromodelling, hobby projects and new media applications. The Arduino can control servo motors, stepper motors, LCD screens, etc. It can take inputs from pressure sensors, accelerometers, GPS modules, gyroscopes, tempera-

ture sensors, humidity sensors, etc. It can interface with XBee (wireless) and bluetooth modules too. Even a basic touchscreen (from the Nintendo DS) can be used for interaction. The screen is available readily and is cheap at close to Rs 700.

The possibilities are immense. What's more is, that you could mash-up and combine sensors to come up with interesting new possibilities. All of this with an easy to use IDE and programming pattern. Examples of what people have done include making an unmanned aerial vehicle (UAV) and weather stations. The limit is purely your imagination. ●

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Related stuff and resources

Arduino has a large community and good support and documentation available on the internet. The resources, references and links are below:

- www.arduino.cc (The Arduino Homepage)
- www.feemo.in (Feemo)
- www.rhydolabz.com (rhydoLABZ)
- <http://www.arduino.cc/en/Main/StandaloneAssembly> (Breadboard Arduino)
- <http://lab.guilhermemartins.net/2009/05/06/paperduino-prints/> (Paperduino—make Arduino on a cardboard)